

ELASTICITY OF DEMAND:

- Elasticity of elasticity? Types
 - price elasticity of demand?
 - Income Elasticity of Demand
 - Cross Elasticity of Demand
- Determinants of Elasticity of Demand
- Application of Elasticity of demand in Business

- So far we've seen that...
 - ❑ Law of demand, when the price rises, the quantity demanded falls.
 - ❑ law of when the price rises, the quantity supplied increases.
 - ❑ Hence, this does not tell us by how much or to what extent the QD of a good will change in response to a change in price.

Elasticity of Demand

- Defined
 - Generally
 - A measure of the responsiveness of the quantity demanded of a good to a change in the price of that good
 - Formally
 - The percentage change in the quantity demanded that results from a percent change in its price

TYPES OF ELASTICITY OF DEMAND

- PRICE ELASTICITY
- INCOME ELASTICITY
- CROSS E ELASTICITY

Computing the Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$E_p = \frac{\frac{\text{change in quantity}}{\text{original quantity}} \times 100}{\frac{\text{change in price}}{\text{original price}} \times 100}$$



- If demand changes a lot or more when price changes-demand is elastic
- If demand changes little, when price changes -inelastic



Example 1:

- Microsoft Office 365.
- Most students need tools like Word, Excel, and PowerPoint for assignments and reports. Even if the price increases slightly, many students will still purchase or use it .

Initial price $P1 = ₹4,000/\text{year}$

New price $P2 = ₹4,400/\text{year}$ (10% increase)

Initial users $Q1 = 1,000$ students

New users $Q2 = 950$ students (5% decrease)

Price elasticity $= -5\%/10\% = -0.5\%$. Demand is inelastic ($|PED| < 1$) — students still buy despite price increase. essential product, less sensitive to price) users did not respond strongly to price drop)



Cross elasticity of demand

- measures responsiveness of demand for one good when the price of another related good changes
- Cross elasticity of demand (E_{xy})
= % Change in Quantity Demanded of Good
/ Change in Price of Good Y
- Examples: (Google Meet vs Zoom)
 - • If $E_{xy} > 0$: Substitutes
 - • If $E_{xy} < 0$: Complements
 - • If $E_{xy} = 0$: Unrelated goods



Example 1

Price of Zoom increases by 10%. Demand for Google Meet rises from 1000 to 1200 users.

% change in price = 10%

% change in quantity = 10

Cross elasticity = 2

→ Positive cross elasticity shows strong substitutes.

→ Positive value confirms that Google Meet and Zoom are substitutes.

provides insights into the relationship between digital platforms like Google Meet and Zoom.

Useful for business strategy, competition, and consumer behavior analysis

Example 2: Price of Google Meet subscription decreases by 5%. Demand for Zoom falls from 800 to 700 users.

Cross elasticity of demand





Introduction to Income Elasticity of Demand

Income Elasticity of Demand (E_y) shows how demand changes when income changes.

- Formula: $E_y = \% \text{ Change in Quantity Demanded} / \% \text{ Change in Income}$
- • $E_y > 1 \rightarrow$ Luxury Good
- • $0 < E_y < 1 \rightarrow$ Normal Good (necessity)
- • $E_y < 0 \rightarrow$ Inferior Good
- Normal goods: positive income elasticity
 - ◆ Necessities: smaller income elasticities
 - ◆ Luxuries: large income elasticities
- Inferior goods: negative income elasticities





Example 1: Productivity Software (MS Office)

Income rises by 10%

Demand rises from 1000 to 1100 licenses

- $\% \Delta m = 10\%$
- $\% \Delta Q = (1100 - 1000) / 1000 \times 100 = 10\%$
- $E_y = 10 / 10 = 1.0$
- $\rightarrow$ Normal good (unitary elasticity)





Determinants of Elasticity of demand

Availability of close substitutes

- more number of substitutes has relatively elastic demand
- with fewer substitutes has relatively inelastic demand

- Nature of the commodity
- Number of uses of a commodity.
- Income level:
- Proportion of income spent on the commodity
- Duration
- Frequency purchased goods is more elastic than rarely purchased goods

